

ECN 594: Two-Part Tariffs and Oligopoly

Nicholas Vreugdenhil

Plan

1. Two-part tariffs: structure and intuition
2. Optimal two-part tariff (homogeneous consumers)
3. Heterogeneous consumers (brief)
4. Cournot competition (refresher)
5. Bertrand competition (refresher)
6. Escaping the Bertrand paradox

Non-linear pricing

- So far: price discrimination by **selection by indicators** and **self-selection**
 - Requires observable characteristics or menu of versions
- Now: **two-part tariffs**
 - Price per unit depends on quantity purchased
 - Consumers *self-select* into how much to buy
- Another form of non-linear pricing

Two-part tariff structure

- A two-part tariff has the form:

$$\text{Total payment} = F + p \times q$$

- F : fixed fee (membership, connection charge)
- p : per-unit price (price per unit consumed)
- q : quantity consumed
- **Examples:**
 - Golf club: membership fee + greens fee per round
 - Phone plan: monthly fee + per-minute charges
 - Costco: annual membership + price per item

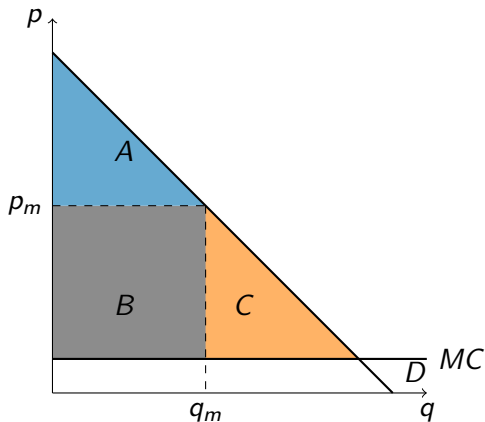
Why is this “non-linear”?

- Average price per unit: $\frac{F+pq}{q} = p + \frac{F}{q}$
- As q increases, average price **decreases**
- This is a quantity discount
- The pricing is “non-linear” because total payment is not proportional to quantity

Plan

1. Two-part tariffs: structure and intuition
2. Optimal two-part tariff (homogeneous consumers)
3. Heterogeneous consumers (brief)
4. Cournot competition (refresher)
5. Bertrand competition (refresher)
6. Escaping the Bertrand paradox

Why use two-part tariffs?



- Under uniform pricing at p_m :
 - CS = Area A
 - Profit = Area B
 - DWL = Area C
- Monopolist “leaves money on the table”
- Can we do better?

Optimal two-part tariff: homogeneous consumers

- Assume all consumers are identical (same demand curve)
- **Key insight:** Use F to extract surplus, use p to maximize total surplus
- **Step 1:** Set $p = MC$
 - This maximizes total surplus (eliminates DWL)
- **Step 2:** Set $F = CS(p = MC)$
 - Extract all consumer surplus through the fixed fee

Why price at marginal cost?

- Firm profit = $CS(p) + (p - c)q(p) = \text{Total surplus}$
- Total surplus is maximized at $p = c$ (no deadweight loss)
- **Intuition:**
 - High p : Earn margin on sales, but reduce CS (and hence F)
 - Low p : Lose margin, but CS increases more
 - Optimal: $p = c$, extract all surplus via F
- **Result:** $p^* = c$, $F^* = CS(c)$, $\pi^* = CS(c)$

Deriving $p^* = c$ with calculus

- For linear demand $q = a - bp$:

$$CS(p) = \frac{(a - bp)^2}{2b}$$

- Profit: $\pi = CS(p) + (p - c)(a - bp)$

- **FOC:**

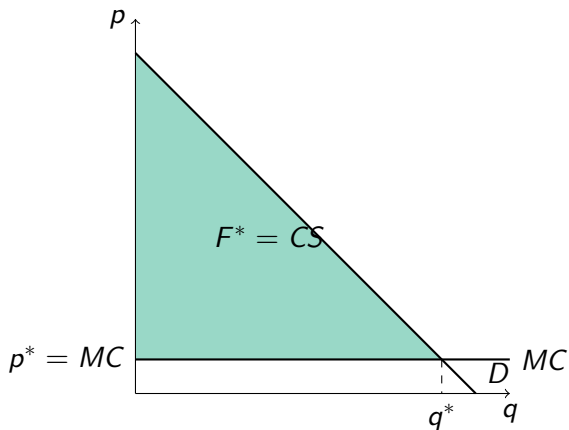
$$\frac{d\pi}{dp} = \frac{dCS}{dp} + (a - bp) - b(p - c) = 0$$

- Since $\frac{dCS}{dp} = -(a - bp)$:

$$-(a - bp) + (a - bp) - b(p - c) = 0$$

- Simplifies to: $p^* = c$

Optimal two-part tariff: graphically



- Set $p^* = MC$
- Consumers buy q^* (efficient quantity)
- Set $F^* =$ entire green triangle
- Profit = $F^* - 0 = F^*$
- **Result:** Firm captures ALL surplus

Worked example: Sushi bar

- **Question:** A sushi bar has $MC = 0.1$ per piece. Each consumer has demand $q = 20 - 10p$.
- (a) What is the optimal uniform price per piece and profit per consumer?
- (b) Consider an all-you-can-eat policy. What is the optimal fee and profit per consumer?
- (c) What is the optimal two-part tariff (fee at the door plus price per piece)?

Worked example: Sushi bar (solution)

- (a) Uniform pricing:

- Inverse demand: $p = 2 - 0.1q$
- Profit: $\pi = (p - 0.1)q = (1.9 - 0.1q)q = 1.9q - 0.1q^2$
- FOC: $\frac{d\pi}{dq} = 1.9 - 0.2q = 0 \Rightarrow q = 9.5$
- Price: $p = 2 - 0.1(9.5) = 1.05$
- Profit: $\pi = (1.05 - 0.1) \times 9.5 = 9.025$

- (b) All-you-can-eat ($p = 0$, charge fee F):

- At $p = 0$: $q = 20$, $CS = \frac{1}{2} \times 20 \times 2 = 20$
- Set $F = 20$. Cost = $0.1 \times 20 = 2$
- Profit: $\pi = 20 - 2 = 18$ (much higher)

Worked example: Sushi bar (solution continued)

- **(c) Optimal two-part tariff:**

- Set $p = MC = 0.1$
- At $p = 0.1$: $q = 20 - 10(0.1) = 19$
- $CS = \frac{1}{2} \times 19 \times (2 - 0.1) = \frac{1}{2} \times 19 \times 1.9 = 18.05$
- Set $F = 18.05$
- Profit from sales: $(0.1 - 0.1) \times 19 = 0$
- Total profit: $\pi = 18.05$

- **Summary:**

- Uniform pricing: $\pi = 9.025$
- All-you-can-eat: $\pi = 18$
- Two-part tariff ($p = MC$): $\pi = 18.05$ (best)

Welfare effects of two-part tariffs

- Compared to uniform monopoly pricing:
- **Producer surplus:** Increases (captures all surplus)
- **Consumer surplus:** Decreases to zero
- **Total surplus:** Increases (DWL eliminated)
- **Efficiency vs equity tradeoff:**
 - More efficient (no DWL)
 - Less equitable (consumers get nothing)

Plan

1. Two-part tariffs: structure and intuition
2. Optimal two-part tariff (homogeneous consumers)
3. Heterogeneous consumers (brief)
4. Cournot competition (refresher)
5. Bertrand competition (refresher)
6. Escaping the Bertrand paradox

What about heterogeneous consumers?

- Reality: consumers have different demand curves
- Problem: if F is too high, some consumers don't participate
- **Tradeoff:**
 - High F : extract more from high-value consumers, but lose low-value ones
 - Low F : serve more consumers, but extract less per consumer
- Optimal F balances these effects

Worked example: Two-part tariff with two types

- **Question:**
- Type H: demand $q_H = 10 - p$, 100 consumers
- Type L: demand $q_L = 5 - p$, 200 consumers
- $MC = 0$
- Compare: (a) $F = 50, p = 0$ vs (b) $F = 12.5, p = 0$

Worked example: Two-part tariff with two types (solution)

- **(a)** $F = 50, p = 0$:
 - Type H: $CS = \frac{1}{2}(10)(10) = 50 \geq F$, joins
 - Type L: $CS = \frac{1}{2}(5)(5) = 12.5 < F$ doesn't join
 - Profit = $50 \times 100 = \$5,000$
- **(b)** $F = 12.5, p = 0$:
 - Type H: $CS = 50 \geq 12.5$, joins
 - Type L: $CS = 12.5 \geq 12.5$, joins
 - Profit = $12.5 \times 300 = \$3,750$
- Better to serve only H (in this case)

Plan

1. Two-part tariffs: structure and intuition
2. Optimal two-part tariff (homogeneous consumers)
3. Heterogeneous consumers (brief)
4. Cournot competition (refresher)
5. Bertrand competition (refresher)
6. Escaping the Bertrand paradox

From ECN 532: Oligopoly models

- You covered Cournot and Bertrand in Hector's class
- Today: quick refresher in IO notation
- **New focus:** Market power measurement
 - Connecting oligopoly models to Lerner index
 - When does each model apply?

Cournot competition: setup

- n firms producing homogeneous goods
- Firms choose **quantities** simultaneously
- Inverse demand: $P = P(Q)$ where $Q = \sum_{i=1}^n q_i$
- Constant marginal cost: c
- Firm i profit: $\pi_i = P(Q) \cdot q_i - c \cdot q_i$

Worked example: Cournot duopoly

- **Setup:** Inverse demand is $P = 100 - Q$. Two firms with $MC = 10$.
- **Steps:**
 1. Write down firm i 's profit function
 2. Find firm i 's best response function
 3. Use symmetry to find equilibrium Q and P

Worked example: Cournot duopoly (solution)

- **Step 1:** Profit: $\pi_i = (100 - q_i - q_j)q_i - 10q_i$
- **Step 2:** FOC: $\frac{d\pi_i}{dq_i} = 100 - 2q_i - q_j - 10 = 0$
- Best response: $q_i = \frac{90 - q_j}{2} = 45 - 0.5q_j$
- **Step 3:** Symmetry: $q_1 = q_2 = q^*$
- $q^* = 45 - 0.5q^* \Rightarrow 1.5q^* = 45 \Rightarrow q^* = 30$
- $Q = 60, P = 100 - 60 = 40$
- Profit per firm: $\pi = (40 - 10) \times 30 = 900$

Cournot with n symmetric firms

- **Setup:**
- n firms with constant marginal cost c
- Linear demand: $P = a - bQ$ where $Q = q_1 + q_2 + \dots + q_n$
- **Question:** What are the Cournot equilibrium quantities and price?

Cournot with n symmetric firms (solution)

- Firm i profit: $\pi_i = q_i(a - bQ) - cq_i$
- FOC: $\frac{d\pi_i}{dq_i} = a - bQ - bq_i - c = 0$
- By symmetry: $q_1 = q_2 = \dots = q_n = q^*$, so $Q = nq^*$
- Substituting: $a - bnq^* - bq^* - c = 0$
- Solving: $q^* = \frac{a-c}{b(n+1)}$, $P^* = \frac{a+nc}{n+1}$
- Profit: $\pi^* = (P^* - c)q^* = \frac{a-c}{n+1} \cdot \frac{a-c}{b(n+1)} = \frac{(a-c)^2}{b(n+1)^2}$
- **Key insight:** As $n \rightarrow \infty$, $P \rightarrow c$ and $\pi \rightarrow 0$

Practice: Cournot comparative statics

- **True, False, or NEI:**
- (a) Adding a third firm to a Cournot duopoly always reduces industry profits.
- (b) In Cournot, all firms must have the same marginal cost.

Practice: Cournot comparative statics (solution)

- **Answers:**
- **(a) TRUE.** More firms $\Rightarrow$ lower price $\Rightarrow$ lower industry profit. Each firm's profit falls, and total profit falls because P is closer to MC .
- **(b) FALSE.** Asymmetric costs work fine. Low-cost firms produce more and earn higher margins.

Plan

1. Two-part tariffs: structure and intuition
2. Optimal two-part tariff (homogeneous consumers)
3. Heterogeneous consumers (brief)
4. **Cournot competition (refresher)**
5. Bertrand competition (refresher)
6. Escaping the Bertrand paradox

Bertrand competition: setup

- n firms producing **homogeneous** goods
- Firms choose **prices** simultaneously
- Consumers buy from lowest-price firm
- If tie: split demand equally
- Constant marginal cost: c

Bertrand: the paradox

- **Nash equilibrium:** $p_1 = p_2 = c$ (marginal cost pricing!)
- **Why?**
 - If $p_i > p_j > c$: firm i can undercut and capture entire market
 - Undercutting continues until $p = c$
- **The “paradox”:**
 - Only 2 firms, but competitive outcome!
 - Zero profits with just 2 competitors
 - Seems unrealistic for most markets

Potential solutions to the Bertrand paradox

- How could we change the assumptions of the Bertrand model to get a more realistic model with positive profits (i.e. firms pricing above MC)?
- **Asymmetric (i.e. different) costs**
 - Like 'cost leadership'
- **Product differentiation/branding**
 - Undercutting the price by a small amount may no longer deliver all of total demand
- **Dynamic competition**
 - What if rivals can retaliate? E.g. you set a high price but threaten to retaliate next time if your rival undercuts you?
- **Capacity constraints**
 - What good is undercutting your rival to get total demand if you cannot actually supply all this demand due to capacity constraints?

Kreps-Scheinkman (1983): resolving the puzzle

- Two-stage game:
 1. Stage 1: Firms choose capacities (quantities)
 2. Stage 2: Firms compete in prices
- **Result:** Equilibrium outcome = Cournot!
- **Intuition:**
 - Capacity choice commits firms
 - Price competition is constrained by capacity
 - Undercutting is limited by what you can produce
- Key insight: commitment matters

When does each model apply?

- **Cournot applies when:**
 - Capacity constraints matter
 - Firms commit to production before selling
 - Quantities are hard to adjust quickly
- Examples: manufacturing, airlines (seat capacity)
- **Bertrand applies when:**
 - Prices adjust quickly
 - No capacity constraints
 - Homogeneous products
- Examples: online retail, commodities

Industry examples: which model?

Industry	Model	Why?
Airlines		

Industry examples: which model?

Industry	Model	Why?
Airlines	Cournot	Capacity (planes, gates) committed

Industry examples: which model?

Industry	Model	Why?
Airlines	Cournot	Capacity (planes, gates) committed
Cement		

Industry examples: which model?

Industry	Model	Why?
Airlines	Cournot	Capacity (planes, gates) committed
Cement	Cournot	Production committed; transport costs

Industry examples: which model?

Industry	Model	Why?
Airlines	Cournot	Capacity (planes, gates) committed
Cement	Cournot	Production committed; transport costs
Gasoline stations		

Industry examples: which model?

Industry	Model	Why?
Airlines	Cournot	Capacity (planes, gates) committed
Cement	Cournot	Production committed; transport costs
Gasoline stations	Bertrand	Prices adjust daily; commodity

Industry examples: which model?

Industry	Model	Why?
Airlines	Cournot	Capacity (planes, gates) committed
Cement	Cournot	Production committed; transport costs
Gasoline stations	Bertrand	Prices adjust daily; commodity
Online retail		

Industry examples: which model?

Industry	Model	Why?
Airlines	Cournot	Capacity (planes, gates) committed
Cement	Cournot	Production committed; transport costs
Gasoline stations	Bertrand	Prices adjust daily; commodity
Online retail	Bertrand	Instant price changes; no capacity

Key Points

1. **Two-part tariff:** $F + p \times q$ (fixed fee + per-unit price)
2. **Optimal (homogeneous):** Set $p = MC$, $F = CS$ at that price
3. Two-part tariffs increase efficiency but reduce consumer surplus to zero
4. **Cournot:** Firms choose quantities; $P > MC$
5. **Bertrand (homogeneous):** $P = MC$, zero profits
6. **Bertrand paradox:** Only 2 firms but competitive outcome
7. Cournot applies with capacity constraints; Bertrand with flexible prices
8. **Escaping the paradox:** Capacity constraints, differentiation, collusion